Cogenetic Minkowski Functional

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Abstract

Following *Cogenesis* [1]: relation is prior to being; spacetime, metrics, and measurement are **projections** of a deeper process-field, not primitives. Minkowski's fusion of time with space into a (3+1) manifold was a **modern calculus utilitarian simplification**—an accounting convention for Lorentz covariance and perturbation theory. Standard practice **reified** that convention into "spacetime." That is the error: fundamentalising time as a fourth dimension on par with space, and treating $(\mathcal{M}, \eta)$ as the relational dynamics of reality rather than as projected bookkeeping.

Cogenetic Minkowski Functional (CMF) is the Minkowski-stage observation functional in that residual-aware framework. It keeps the calculus machinery—flat η, light cones, variational methods—while refusing the reification. η is the **zeroth-order low-residual projection**, not ground truth. Environment enters as C—the environmental density gradient along the observer's throughline. **Process time** t indexes resolution $R(t)$—not a rename of t as resolution, but the map $t \mapsto R(t)$ at which the functional is evaluated and sharpens as the process runs.

CMF is a **refinement algorithm** on the **projection calculus**—against the **process calculus** beneath it: project, measure residual, recover flat η only where $|r_{C,R}|$ is negligible. Flat Minkowski remains *usable* when the utilitarian simplification suffices—when ignoring C and $R(t)$ is adequate for the accounting task, not because spacetime is true.

This document specifies the construction, contrasts it with the bolted-on input pattern of standard practice, and separates what Minkowski *stipulates* from what CMF *determines* at each process time and *predicts* over the running observation. A reference implementation with a satellite point-detector example is included.

1 Problem statement

1.1 What the Minkowski calculus is for

On $\mathcal{M} = \mathbb{R}^{3,1}$ with metric $\eta_{\mu\nu} = \mathrm{diag}(-c^2, 1, 1, 1)$, the interval

$$ds^2 = \eta_{\mu\nu}\, dx^\mu dx^\nu \tag{1}$$

is a **utilitarian simplification**: a flat stage for special relativity, QFT regularisation, and linearised gravity. The apparatus is proven and cheap to use. None of that makes $(\mathcal{M}, \eta)$ a reification of relational reality.

1.2 Where reification goes wrong

Minkowski's (3+1) fusion was an accounting hack—time promoted to a fourth coordinate so Lorentz transformations act linearly on a single object. Standard practice then:

1. fixes a functional early—e.g. $F[\Phi] = \int |\Phi|^2 \, d\mu$, an action $S[\Phi]$, or a Green's function;
2. treats *environment* and *process* as *external inputs* bolted on afterward;
3. treats *events* as primitive points on $\mathcal{M}$;
4. identifies this *calculus projection* with *physical reality.*

Spacetime as primitive is a category error. It hides what any finite observation already carries: observer location, environment along the throughline, running process time t and resolution $R(t)$, and the residual lost when a continuous field is projected to an observable. CMF makes these explicit *inside* the functional, not as afterthoughts.

1.3 Design principle

Process calculus carries the relational dynamics: observer throughline, environment C, process time t, resolution $R(t)$, and the residual $r_{C,R}$ lost on projection.

Projection calculus is the utilitarian simplification Minkowski built from that dynamics: flat $(\mathcal{M}, \eta)$, intervals, Lorentz covariance, variational machinery [1]. It is bookkeeping that converges when $|r_{C,R}|$ is negligible—not a second ontology.

CMF uses both without collapsing them:

$$\boxed{\text{Process calculus is primary; } (\mathcal{M}, \eta) \text{ is projection calculus, not relational reality.}} \tag{2}$$

2 Formal construction

2.1 Base stage

Retain the Minkowski background

$$\mathcal{M} = \mathbb{R}^{3,1}, \qquad \eta_{\mu\nu} = \mathrm{diag}(-c^2, 1, 1, 1) \tag{3}$$

as the working stage of **projection calculus** (§1.3)—not **process calculus**.

2.2 Process field

Introduce a smooth process field

$$\Phi : \mathcal{M} \to V \tag{4}$$

where V is a finite-dimensional vector space (scalar, spinor, or internal coherence space).

2.3 Context C

C is the declared local condition under which process becomes distinguishable:

$$C = (U,\, e_a^{\mu},\, A,\, B,\, I,\, E,\, G) \tag{5}$$

Operational form (primary). For a point detector on a throughline—e.g. a satellite sighting a surface target—C is the environmental density gradient along the observer ray:

$$C(x) = \hat{n} \cdot \nabla \rho_{\mathrm{env}}(x) \tag{6}$$

where ρ_{env} is environmental density (atmosphere, medium, noise background) and $\hat{n}$ is the unit vector from observer to target. C is not metadata bolted onto a fixed functional; it shapes the map itself.

Table 1: Components of context C.

Component	Meaning
$U \subset \mathcal{M}$	Region of distinguishability
e^{μ}_{a}	Observer / tetrad frame
A	Apparatus or observation functional
B	Boundary / initial data
I	Preparation / input condition
E	Environmental coupling
G	Gauge / coordinate convention

2.4 Process time and resolution $R(t)$

In the core formalism, R denotes **resolution**—finite distinction capacity—not calendar time. CMF does *not* identify R with coordinate time t. Instead, **process time** t is the running index of the observation process, and t maps to an evolving resolution state:

$$t \longmapsto R(t) = \big(\ell(t),\, \tau(t),\, \Lambda(t),\, \epsilon(t),\, \Sigma(t)\big) \tag{7}$$

At each t, the map uses $R(t)$ inside $\Pi_{C,R(t)}$, $|r_{C,R(t)}|$, and $O(C, R(t))$. Early t: coarse (ℓ, τ, Λ). As the process runs, $R(t)$ sharpens. t is *when* the observation is happening; $R(t)$ is *how sharply* it can distinguish at that moment.

Table 2: Components of resolution $R(t)$ at process time t.

Component	Meaning
$\ell(t)$	Spatial resolution
$\tau(t)$	Temporal sampling window
$\Lambda(t)$	Bandwidth cutoff
$\epsilon(t)$	Distinction tolerance
$\Sigma(t)$	Noise / error covariance

(ℓ, τ, Λ) improve as t advances; (ϵ, Σ) carry apparatus and environmental limits at each t. Subscripts on C, R mean evaluation at the resolution state $R(t)$ reached at process time t.

2.5 Projection operator

$$(\Pi_{C,R}\Phi)(x) = \int_U K_{C,R}(x, y)\, A_C[\Phi](y)\, d\mu_\eta(y) \tag{8}$$

where $K_{C,R}$ is a smooth sampling kernel determined by $(\ell, \tau, \Lambda, \epsilon, \Sigma)$ and weighted by context C along the throughline, and A_C is the apparatus functional.

2.6 Residual

$$r_{C,R}[\Phi] = \Phi - \iota_C \Pi_{C,R}\Phi \tag{9}$$

where ι_C embeds the projected observable back into the modelling space. The residual is **measurable**:

$$|r_{C,R}|^2 = \int_U \langle r_{C,R}(x), r_{C,R}(x)\rangle\, d\mu_\eta(x) \tag{10}$$

This is exactly the information a pure Minkowski projection suppresses.

3 Effective metric and interval

3.1 Metric family

$$g_{\mu\nu}^{C,R} = \eta_{\mu\nu} + h_{\mu\nu}^{C,R}[\Pi_{C,R}\Phi,\, r_{C,R}] \tag{11}$$

First-order calculus-compatible correction:

$$h_{\mu\nu}^{C,R} = a\, \partial_\mu \rho_{C,R}\, \partial_\nu \rho_{C,R} + b\, J_\mu^r J_\nu^r + c\, \Xi_{\mu\nu}^{C,R} \tag{12}$$

where

$$\rho_{C,R} = |\Pi_{C,R}\Phi|^2, \qquad J_\mu^r = \partial_\mu |r_{C,R}|^2 \tag{13}$$

and $\Xi_{\mu\nu}^{C,R}$ is a fragmentation / decoherence tensor from projected gradients and noise.

3.2 Interval

$$ds_{C,R}^2 = g_{\mu\nu}^{C,R}\, dx^\mu dx^\nu = \eta_{\mu\nu}\, dx^\mu dx^\nu + h_{\mu\nu}^{C,R}\, dx^\mu dx^\nu \tag{14}$$

3.3 Minkowski recovery condition

Minkowski is recovered when

$$\Pi_{C,R}\Phi \text{ is uniform}, \qquad r_{C,R} \to 0, \qquad \Xi_{\mu\nu}^{C,R} \to 0. \tag{15}$$

Equivalently: **flat η is the (C,R)-low-residual limit of the projected functional—a simplification that converges, not a discovered spacetime.**

4 The refinement algorithm

CMF is not parameter tuning. It is a **process**:

1. **Declare** C—fix the environment and observer throughline (e.g. satellite position, target, ρ_{env}).
2. **Set process time** t—resolve $R(t) = (\ell(t), \tau(t), \Lambda(t), \epsilon(t), \Sigma(t))$ and evaluate at this moment of observation.
3. **Project**—compute $\Pi_{C,R(t)}\Phi$.
4. **Measure residual**—compute $|r_{C,R(t)}|^2$.
5. **Advance process time**—as t increases, $R(t)$ sharpens and the functional refines.
6. **Recover Minkowski**—when residual falls below tolerance, flat geometry is a valid zeroth-order description.

$$\boxed{(\mathcal{M}, \eta) \;\longrightarrow\; (\mathcal{M}, \eta, \Phi, \Pi_{C,R}, r_{C,R}, g^{C,R})} \tag{16}$$

5 Observers and events

Events are **not** primitive points. An observation is

$$O_i(C,R) = \int_U K_{C,R}^i(x)\, F_i(\Phi, \partial\Phi, g^{C,R})\, d\mu_{g^{C,R}} + \epsilon_i \tag{17}$$

a finite-resolution projection of field-process into an observable, with sampling noise ϵ_i.

Table 3: Standard practice vs. CMF.

Standard	CMF
Fixed $F[\Phi]$	Process functional $F[\Phi; C, R(t)]$
C, R bolted on as inputs	C, R inside the map
Same answer at all times	Functional sharpens with process time
Event = point on $\mathcal{M}$	Event = kernel projection

6 Dynamics (optional extension)

Field dynamics remain calculus-based. Action:

$$S[\Phi] = \int_{\mathcal{M}} \left[\tfrac{1}{2}\,\eta^{\mu\nu}\langle\partial_\mu\Phi, \partial_\nu\Phi\rangle - V(\rho_{C,R}, \Xi) + \tfrac{\alpha}{2}\,|r_{C,R}|^2\right] d^4x \tag{18}$$

Euler–Lagrange equations yield PDEs with residual source terms:

$$\Box_\eta \Phi + \frac{\partial V}{\partial \Phi} = J_r \tag{19}$$

The residual source J_r is what old Minkowski projection would drop.

7 When the utilitarian simplification suffices

Use flat η when the accounting simplification is adequate for the task—not because spacetime is true.

Table 4: When flat η suffices vs. when to carry CMF.

Situation	Recommendation
Back-of-envelope kinematics	Flat η—simplification suffices
Lab frame, long integration, buried error budget	Often suffices
Satellite line-of-sight through density gradient	Carry CMF
Early integration, targeting decisions	Carry CMF
Residual-dominated regime	Do not reify η as ontology

Flat Minkowski is a **utilitarian simplification**, not relational reality. CMF adds no overhead when the simplification is adequate: set $h \to 0$ and recover η. Carry C and $R(t)$ when the accounting task requires what the projection suppresses.

8 Stipulations vs. determination

CMF is a **cogenetic** projection formalism [1]: observables co-emerge with context and process time. Judging it against general relativity or quantum field theory as a "correction" is the wrong frame—those theories already presuppose flat η as stage [2]. The correct comparison is **CMF vs. standard Minkowski practice**.

The distinction is methodological, not a claim that CMF never predicts anything:

- **Minkowski stipulates**—at a given setup, standard practice *fixes* $|r_{C,R}| = 0$, $g = \eta$, and a functional with no R-map inside it. These are prior commitments of the calculus stage, not outputs of an observation process.

- **CMF determines**—at each process time t, the map resolves $R(t)$ and computes $|r_{C,R(t)}|$, $g^{C,R(t)} = \eta + h$, and $O(C, R(t))$ from (Φ, C, t) via project $\to$ measure residual $\to$ evaluate observer (17).
- **CMF predicts**—over process time, the formalism claims refinement behaviour: falling residual, sharpening $O(C, R)$, and a bolted-on $F[\Phi]$ that cannot track any of it.

8.1 What Minkowski stipulates

Table 5: Values fixed by standard Minkowski practice before the observation process runs.

Stipulation	Value
Residual	$\lvert r_{C,R}\rvert = 0$
Metric	$g^{\mu\nu} = \eta^{\mu\nu}$
Interval correction	$\lvert g_{xx} - \eta_{xx}\rvert = 0$
Functional	$F[\Phi]$ with no R-dependence
Over process time	No change—C and R bolted on externally

These are utilitarian defaults—often adequate. They are not computed from the throughline environment or the running observation.

8.2 What CMF determines at each R

At every process time, CMF **derives** quantities Minkowski practice cannot represent:

- **Residual** $|r_{C,R}|^2 = |\Phi - \iota_C \Pi_{C,R}\Phi|^2$ along the throughline.
- **Metric correction** $h[\Pi\Phi, r]$ and $g^{C,R} = \eta + h$; in vacuum $g_{xx} = 1$, where $|C| > 0$ and the process field has structure, $|g_{xx} - \eta_{xx}|$ is generally nonzero.
- **Observer reading** $O(C, R)$ from (17)—a kernel projection of $\rho_{C,R} = |\Pi_{C,R}\Phi|^2$ under $d\mu_{g^{C,R}}$, not raw $|\Phi|^2$ at a point.

What we do *not* compare. CMF does **not** claim that $O(C, R)$ converges to raw $|\Phi|^2$ at a point. Comparing $O(C, R)$ to $|\Phi(0, 0)|^2$ is the wrong metric. The correct comparison is **stipulation vs. determination**: what Minkowski *fixes in advance* versus what CMF *computes from the map* at each R.

8.3 What CMF predicts over R

Table 6: Process behaviour: stipulation vs. determination vs. prediction.

Minkowski (stipulates)	CMF (determines / predicts)
$\lvert r_{C,R}\rvert = 0$ always	determines $\lvert r_{C,R}\rvert^2$; coarse-to-fine refinement drives it down
O independent of R	determines $O(C, R(t))$; predicts sharpening with R
$g^{\mu\nu} = \eta^{\mu\nu}$	determines $h[\Pi\Phi, r] \neq 0$ where $\lvert C\rvert > 0$
$\lvert g_{xx} - \eta_{xx}\rvert = 0$	determines $\lvert g_{xx} - \eta_{xx}\rvert$ at target
No R-map inside $F[\Phi]$	predicts $\lvert O(C, R) - O(C, R_{\text{late}})\rvert \to 0$ as process runs

Testable claims.

P1: Residual refinement *(predicts).* For fixed C and Φ, coarse-to-fine process time drives $|r_{C,R}|^2$ down over the main observation trajectory. Very late R may show a local uptick when projection sharpens on a structured target—the claim is refinement, not strict monotonicity at every step.

P2: Functional dependence on R *(predicts).* $F[\Phi; C, R(t)]$ and $O(C, R)$ vary with R; a bolted-on $F[\Phi] = \int |\Phi|^2 \, d\mu$ does not.

P3: Context-driven metric correction *(determines).* $h[\Pi\Phi, r]$ correlates with throughline context $|C|$, not identically zero as η stipulates.

P4: Interval correction *(determines).* $|g_{xx} - \eta_{xx}|$ at the target is computed by CMF; Minkowski stipulates zero. In vacuum, $g_{xx} = 1$ throughout; at the structured target the correction need not shrink with R—the point is that CMF *represents* it.

P5: Process refinement *(predicts).* $|O(C, R) - O(C, R_{\text{late}})|$ decreases with R. Standard Minkowski practice offers no R-dependence inside $F[\Phi]$; a bolted-on fixed integral is unchanged at all process times.

Verification (satellite demo). The reference implementation separates the three roles on the worked setup below. **Stipulations:** Minkowski fixes $|r| = 0$, $|g_{xx} - 1| = 0$, and $F[\Phi] \approx 4.52$ at all R. **Determinations:** at early R, CMF computes $|r_{C,R}|^2 \approx 3.1$ and $|g_{xx} - 1| \approx 0.06$; at late R, $|r_{C,R}|^2 \approx 0.05$ and $|g_{xx} - 1| \approx 2.4$ at the target—nonzero structure Minkowski cannot represent. **Predictions:** residual falls from early to late R (with a modest uptick in the final refinement steps on this grid); $|O(C, R) - O(C, R_{\text{late}})|$ approaches zero; the bolted-on functional is unchanged throughout.

9 Worked example: satellite point detector

Setup.

- Observer (satellite) at altitude $x = -7$, target at surface $x = 0$.
- Environmental density: $\rho_{\text{env}}(x) = \exp(-|x|/H)$ (exponential atmosphere).
- Context: $C = \hat{n} \cdot \nabla \rho_{\text{env}}$ along the throughline (6).
- Process field Φ: Gaussian target packet on the surface.

Results (reference implementation). At early process time ($R = t = -4$), projection is coarse: $|r_{C,R}|^2 \approx 3.1$ and $g_{xx} \approx 1.06$ at the target (cf. Minkowski $\eta_{xx} = 1$). At late process time ($R = t = 4$), the residual falls to $|r_{C,R}|^2 \approx 0.05$; refinement error $|O(C, R) - O(C, R_{\text{late}})|$ approaches zero. The process functional $F[\Phi; C, R(t)]$ shifts from ≈ 0.002 to ≈ 0.58 while a fixed functional $F[\Phi] = \int |\Phi|^2 \, d\mu \approx 4.52$ is unchanged at all R—C and R play no role inside the bolted-on map. Vacuum regions carry $g_{xx} = 1$ throughout; flat Minkowski is the valid zeroth-order description where $|r_{C,R}|$ is negligible, not necessarily where the resolved process field has structure.

10 Reference implementation

11 Relation to deeper process geometry

The *Cogenesis* construction [1] gives a deeper form:

$$d\Sigma^2_{C,R} = \langle \delta_{C,R}\Phi,\, G_{C,R}[\Phi]\, \delta_{C,R}\Phi \rangle + r_\Sigma \tag{20}$$

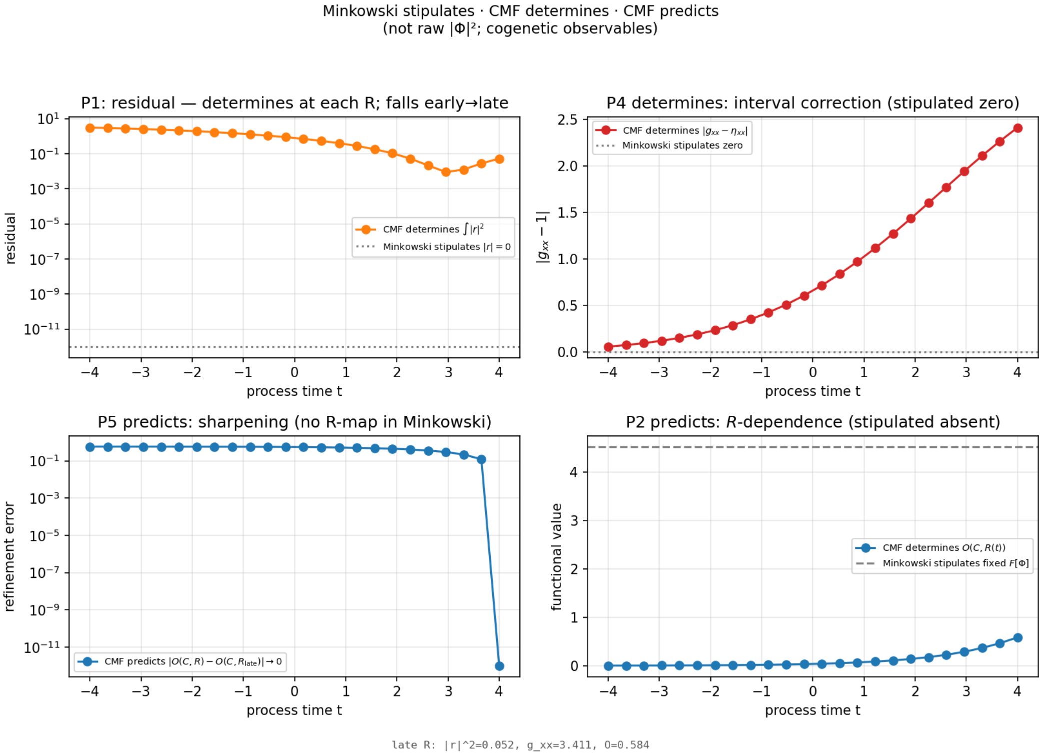
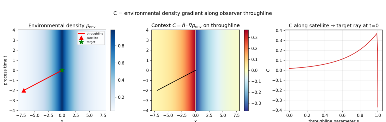
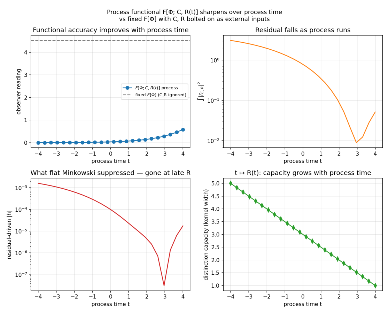


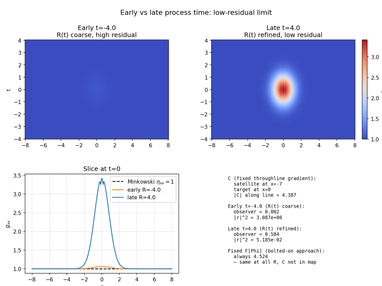
Figure 1: Stipulations vs. determination vs. prediction (`python demo.py`). Top row: CMF *determines* residual and interval correction; Minkowski *stipulates* both zero. Bottom row: CMF *predicts* refinement and R-dependence; Minkowski *stipulates* no R-map.



(a) Context C along throughline



(b) Process vs. bolted-on functional



(c) Early vs. late process time

Figure 2: Reference implementation figures (`python demo.py`).

Table 7: Release artefacts.

File	Role
`cmf.tex`	LaTeX source (this document)
`cmf.pdf`	Compiled release PDF
`model.py`	Core CMF: C as throughline gradient; $t \mapsto R(t)$ resolution state
`demo.py`	Satellite demonstration; generates release figures
`requirements.txt`	`numpy`, `scipy`, `matplotlib`
`README.md`, `LICENSE`	Release metadata
`output/`	Demonstration figures

with classical metric geometry recovered by projection:

$$ds^2 = \pi_{\text{geom}}(d\Sigma^2_{C,R}) \tag{21}$$

CMF represents this projection as an effective metric family $g^{C,R}_{\mu\nu}$—calculus-safe, without pretending the full $d\Sigma^2$ has been captured by ds^2 alone. Where a single shared causal cone and metric already unify sectors at the calculus level [2], CMF specifies how flat η arises as the low-residual projection of that deeper process under declared context and process time.

12 Conclusion

CMF keeps the Minkowski projection calculus without reifying it—against the process calculus that carries C, $R(t)$, and residual:

- **Same machinery**—manifolds, tensors, kernels, PDEs, variational methods.
- **Same simplification exit**—flat η when $|r_{C,R}|$ is negligible and the accounting task allows it.
- **No reification**—C and $R(t)$ inside the functional; process time indexes resolution, not a fused spacetime ontology.
- **Relational dynamics**—events, intervals, and metrics from observation process, not primitive points on $\mathcal{M}$.

The upgrade is not heavier machinery. It is refusing to mistake projected bookkeeping for ground.

Reified:	$(\mathcal{M}, \eta_{\mu\nu})$	as primitive spacetime
CMF:	$(\mathcal{M}, \eta_{\mu\nu}, \Phi, \Pi_{C,R}, r_{C,R}, g^{C,R}_{\mu\nu})$	
		flat η when $\lvert r_{C,R}\rvert \to 0$—utilitarian simplification, not ontology.

References

[1] M. Petrik and Militant.AI, *Cogenesis*, Zenodo, 2026. doi:10.5281/zenodo.20539999

[2] M. Petrik and Militant.AI, *One-Metric Quantum Gravity: Unifying SR, GR, and QM with a Shared Causal Cone*, Zenodo, 2025. doi:10.5281/zenodo.17731104